

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING****Product Identifier**

Perihal Produk: **EMB Agar, Modified**  
Product Description: **EMB Agar, Modified**  
Cat No. : R01405, R01406

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

**Details of the supplier of the safety data sheet**

Company Thermo Scientific Microbiology Sdn Bhd  
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Cheng, 75250 Melaka, Malaysia  
+606 334 0975 .

Supplier Remel  
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**Emergency Telephone Number**

(603) 5122 8888  
CHEMTREC Malaysia **1-800-815-308** (Malay)  
CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION****Classification of the substance or mixture****Label Elements****Hazard Statements****Precautionary Statements****Other Hazards**

This product does not contain any known or suspected endocrine disruptors

**SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

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| Component                           | CAS No     | Weight % |
|-------------------------------------|------------|----------|
| Yeast, ext.                         | 8013-01-2  | Trace    |
| Gelatins, hydrolyzates              | 68410-45-7 | 0.34     |
| Acid red 87                         | 17372-87-1 | Trace    |
| Methylene blue trihydrate           | 7220-79-3  | Trace    |
| Sodium carbonate                    | 497-19-8   | Trace    |
| Agar                                | 9002-18-0  | 1.1      |
| Dihydrogen potassium phosphate      | 7778-77-0  | Trace    |
| Propanoic acid, 2-oxo-, sodium salt | 113-24-6   | Trace    |
| .beta.-D-Lactose                    | 63-42-3    | 0.77     |
| Sucrose                             | 57-50-1    | 0.05     |
| Dipotassium phosphate               | 7758-11-4  | Trace    |
| Water                               | 7732-18-5  | 96.53    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

**Eye Contact** Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact** Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Ingestion** Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Inhalation** Remove to fresh air. Get medical attention immediately if symptoms occur.

**Self-Protection of the First Aider** No special precautions required.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. . Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

### Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

None under normal use conditions.

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## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | Malaysia | ACGIH TLV                 | OSHA PEL                                                                                                                           |
|-----------|----------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Sucrose   |          | TWA: 10 mg/m <sup>3</sup> | (Vacated) TWA: 15 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> |

| Component | European Union | The United Kingdom                                                  | Germany |
|-----------|----------------|---------------------------------------------------------------------|---------|
| Sucrose   |                | STEL: 20 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr |         |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

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## Personal protective equipment

|                                 |                                                    |
|---------------------------------|----------------------------------------------------|
| <b>Eye Protection</b>           | Wear safety glasses with side shields (or goggles) |
| <b>Hand Protection</b>          | Protective gloves                                  |
| <b>Skin and body protection</b> | Long sleeved clothing                              |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                               |
|---------------------------------|---------------------------------------------------------------|
| <b>Respiratory Protection</b>   | No protective equipment is needed under normal use conditions |
| <b>Recommended Filter type:</b> | Particle filter                                               |

|                                |                                                                       |
|--------------------------------|-----------------------------------------------------------------------|
| <b><u>Hygiene Measures</u></b> | Handle in accordance with good industrial hygiene and safety practice |
|--------------------------------|-----------------------------------------------------------------------|

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b><u>Environmental exposure controls</u></b> | No information available |
|-----------------------------------------------|--------------------------|

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### **Appearance**

|                       |                          |
|-----------------------|--------------------------|
| <b>Physical State</b> | Solid Gel Consistency    |
| <b>Odor</b>           | No information available |
| <b>Odor Threshold</b> | No data available        |
| <b>pH</b>             | No information available |

|                            |                          |                                          |
|----------------------------|--------------------------|------------------------------------------|
| <b>Melting Point/Range</b> | No data available        |                                          |
| <b>Softening Point</b>     | No data available        |                                          |
| <b>Boiling Point/Range</b> | No information available |                                          |
| <b>Flash Point</b>         | No information available | <b>Method -</b> No information available |

|                                 |                          |
|---------------------------------|--------------------------|
| <b>Evaporation Rate</b>         | No data available        |
| <b>Flammability (solid,gas)</b> | No information available |
| <b>Explosion Limits</b>         | No data available        |

|                                     |                          |             |
|-------------------------------------|--------------------------|-------------|
| <b>Vapor Pressure</b>               | No data available        |             |
| <b>Vapor Density</b>                | No data available        | (Air = 1.0) |
| <b>Specific Gravity / Density</b>   | No data available        |             |
| <b>Bulk Density</b>                 | No data available        |             |
| <b>Water Solubility</b>             | No information available |             |
| <b>Solubility in other solvents</b> | No information available |             |

#### **Partition Coefficient (n-octanol/water)**

|                  |                |
|------------------|----------------|
| <b>Component</b> | <b>log Pow</b> |
| Acid red 87      | 4.80           |
| Sucrose          | -3.67          |

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|                           |                          |
|---------------------------|--------------------------|
| Autoignition Temperature  | No data available        |
| Decomposition Temperature | No data available        |
| Viscosity                 | No data available        |
| Explosive Properties      | No information available |
| Oxidizing Properties      | No information available |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### Conditions to Avoid

None known.

### Incompatible Materials

None known.

### Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Acute Toxicity

#### Toxicology data for the components

| Component                           | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation              |
|-------------------------------------|---------------------------|------------------------------|------------------------------|
| Acid red 87                         |                           | LD50 > 2000 mg/kg ( Rat )    |                              |
| Sodium carbonate                    | 2800 mg/kg ( Rat )        | > 2000 mg/kg (rabbit)        | 2.3 mg/l 2h (Rat)            |
| Agar                                | LD50 = 11 g/kg ( Rat )    |                              |                              |
| Dihydrogen potassium phosphate      | LD50 = 3200 mg/kg ( Rat ) | LD50 > 4640 mg/kg ( Rabbit ) | LC50 > 0.83 mg/L ( Rat ) 4 h |
| Propanoic acid, 2-oxo-, sodium salt | 5600 mg/kg (Rat)          |                              |                              |

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|                       |                            |                              |   |
|-----------------------|----------------------------|------------------------------|---|
| .beta.-D-Lactose      | LD50 > 10 g/kg ( Rat )     |                              |   |
| Sucrose               | LD50 = 29700 mg/kg ( Rat ) |                              |   |
| Dipotassium phosphate | 8 g/kg (rat)               | LD50 > 5000 mg/kg ( Rabbit ) |   |
| Water                 | -                          | -                            | - |

## Chronic Toxicity

### Carcinogenicity

There are no known carcinogenic chemicals in this product

### Legend:

X - Listed 'I' - Not Listed XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B))

### Sensitization

No information available

### Mutagenic Effects

No information available

### Reproductive Effects

No information available

### Developmental Effects

No information available

### Target Organs

No information available.

### Symptoms

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component        | Freshwater Fish                                                                       | Water Flea                               | Freshwater Algae | Microtox |
|------------------|---------------------------------------------------------------------------------------|------------------------------------------|------------------|----------|
| Acid red 87      | LC50= 1200 mg/L/48h<br>(Oryzias latipes)                                              |                                          |                  |          |
| Sodium carbonate | Lepomis macrochirus:<br>LC50: 300 mg/L/96h<br>Gambusia affinis: LC50:<br>740 mg/L/96h | EC50: = 265 mg/L, 48h<br>(Daphnia magna) |                  | -        |

### Persistence and degradability

No information available

### Bioaccumulative potential

No information available

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Acid red 87 | 4.80    | No data available             |
| Sucrose     | -3.67   | No data available             |

### Mobility in soil

No information available.

### Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### Waste from Residues/Unused

Dispose of in accordance with local regulations

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## Products

### Contaminated Packaging

Empty containers should be taken to an approved waste handling site for recycling or disposal

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

### Road and Rail Transport

Not regulated

### IATA

Not regulated

### Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

X = listed

| Component                           | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL       |
|-------------------------------------|-----------|------|-----|-------|------|------|-------|------|------------|
| Yeast, ext.                         | -         | X    | X   | X     | -    |      | X     | X    | KE-05-1355 |
| Gelatins, hydrolyzates              | -         | X    | X   | X     | -    |      | X     | X    | KE-17576   |
| Acid red 87                         | 241-409-6 | X    | X   | X     | X    | X    | X     | X    | KE-06722   |
| Sodium carbonate                    | -         | X    | X   | X     | X    | X    | X     | X    | KE-31380   |
| Agar                                | -         | X    | X   | X     | -    |      | X     | X    | KE-00275   |
| Dihydrogen potassium phosphate      | -         | X    | X   | X     | X    | X    | X     | X    | KE-28622   |
| Propanoic acid, 2-oxo-, sodium salt | 204-024-4 | X    | X   | X     | X    | X    | X     | X    | KE-27653   |
| .beta.-D-Lactose                    | -         | X    | X   | X     | -    | X    | X     | X    | KE-17416   |
| Sucrose                             | -         | X    | X   | X     | -    | X    | X     | X    | KE-17258   |
| Dipotassium phosphate               | -         | X    | X   | X     | X    | X    | X     | X    | KE-12167   |
| Water                               | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400   |

### National Regulations

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

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**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances  
**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date** 29-Mar-2023  
**Revision Summary** Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**